



Optimal PHASES Scoring for Risk Stratification of Surgically Treated Unruptured Aneurysms

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■ **OBJECTIVE:** The PHASES (Population, Hypertension, Age, Size, Earlier subarachnoid hemorrhage, Site) score was developed to facilitate risk stratification for management of unruptured intracranial aneurysms (UIAs). This study aimed to identify the optimal PHASES score cutoff for predicting neurologic outcomes in patients with surgically treated aneurysms.

■ **METHODS:** All patients who underwent micro-neurosurgical treatment for UIA at a large quaternary center from January 1, 2014, to December 31, 2020, were retrospectively reviewed. Inclusion criteria included a modified Rankin Scale (mRS) score of ≤ 2 at admission. The primary outcome was 1-year mRS score, with a “poor” neurologic outcome defined as an mRS score > 2 .

■ **RESULTS:** In total, 375 patients were included in the analysis. The mean (SD) PHASES score for the entire study population was 4.47 (2.67). Of 375 patients, 116 (31%) had a PHASES score ≥ 6 , which was found to maximize prediction of poor neurologic outcome. Patients with PHASES scores ≥ 6 had significantly higher rates of poor neurologic outcome than patients with PHASES scores < 6 at discharge (58 [50%] vs. 90 [35%], $P = 0.005$) and follow-up (20 [17%] vs. 18 [6.9%], $P = 0.002$). After adjusting for age, Charlson Comorbidity Index score, nonsaccular aneurysm, and aneurysm size, PHASES score ≥ 6 remained a significant predictor of poor neurologic outcome at follow-up (odds ratio, 2.75; 95% confidence interval, 1.42–5.36, $P = 0.003$).

■ **CONCLUSIONS:** In this retrospective analysis, a PHASES score ≥ 6 was associated with significantly greater proportions of poor outcome, suggesting that awareness of this threshold in PHASES scoring could be useful in risk stratification and UIA management.

INTRODUCTION

The current worldwide prevalence of intracranial aneurysms is estimated to be approximately 3%.^{1–3} Diagnostic imaging has revealed an increasing number of unruptured intracranial aneurysms (UIAs),^{4,5} but whether incidentally discovered UIAs should be treated remains unclear. Currently, subarachnoid hemorrhage (SAH) after aneurysm rupture is associated with mortality rates ranging from 35% to 50%.³ Even if a patient survives aneurysm rupture, there is a high likelihood of permanent disability, with morbidity rates approaching 60%.⁶ These dangers suggest the value of prophylactically treating UIAs. However, both endovascular and open surgical treatments pose a nonnegligible risk for patients, with a combined mortality and morbidity of approximately 5% for each treatment.⁷ Because many aneurysms may never rupture, physicians must thoughtfully weigh the risk of rupture against the risks associated with treatment to ensure optimal patient outcomes. Accurately determining the risk of aneurysm rupture plays a crucial role in deciding between observation and treatment.

On the basis of data from 6 prospective cohort studies that assessed 8283 patients and 29,166 years of patient follow-up,

Key words

- Outcomes
- PHASES
- Risk stratification
- Scoring
- Unruptured aneurysm

Abbreviations and Acronyms

CCI: Charlson Comorbidity Index

mRS: modified Rankin Scale

PHASES: Population, Hypertension, Age, Size, Earlier subarachnoid hemorrhage, Site

SAH: Subarachnoid hemorrhage

UIA: Unruptured intracranial aneurysm

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Greving et al.⁸ developed a practical risk-scoring system called the PHASES (Population, Hypertension, Age, Size, Earlier subarachnoid hemorrhage, Site) score. Released in 2014, the PHASES score uses routinely assessed patient and aneurysm characteristics to determine the 5-year risk of aneurysm rupture. Because of its use in decision-making regarding the clinical treatment of unruptured aneurysms,⁹ the PHASES score has been criticized for not taking into account the risks associated with surgical aneurysm treatment.^{10,11} In an effort to elucidate the association between the PHASES score and treatment risk, we sought to evaluate whether the PHASES score is predictive of neurologic outcome in patients with surgically treated UIAs.

METHODS

All patients who underwent surgical treatment for a UIA between January 1, 2014, and December 31, 2020, at a single quaternary center (Barrow Neurological Institute, St. Joseph's Hospital and Medical Center, Phoenix, Arizona) were identified in a retrospective research database. Inclusion criteria were the availability of treatment data, adequate follow-up, and a modified Rankin Scale (mRS) score of ≤ 2 at admission. Adequate follow-up was defined as at least 1 in-office visit more than 7 days from the index hospitalization related to the aneurysm operation. Patients were grouped into 2 cohorts based on PHASES scoring (≥ 6 vs. < 6).⁸ PHASES scores were calculated for all patients; for patients with multiple aneurysms, the highest PHASES score was used. The primary outcome was 1-year mRS score, with a poor neurologic outcome defined as an mRS score of > 2 .

This study was approved by the Institutional Review Board for Human Research at St. Joseph's Hospital and Medical Center in Phoenix and complied with the Health Insurance Portability and Accountability Act. The need for patient consent was waived by the institutional review board because of the retrospective nature of the study. Electronic medical records were analyzed for demographic information (age, sex, and comorbidities), American Society of Anesthesiologists classification, Charlson Comorbidity Index (CCI) score, and life expectancy as determined by CCI score. Aneurysm characteristics obtained from computed tomography or angiogram findings were collected to assess for aneurysm maximum diameter, neck diameter, height perpendicular, maximum height, calcification, and type; critical perforating or branch vessels; intraluminal thrombosis; and size and aspect ratio.

Statistical analyses included data aggregation, exploratory analysis, and multivariate analysis using R, version 4.0.1 (R Foundation for Statistical Computing). Optimal cutoff points for continuous variables were determined using Euclidean distance analysis onto mRS score > 2 at final follow-up. Univariate analysis was used to initially determine variable association with outcome. All variables with a P value < 0.2 on univariate analysis were included in a multivariable logistic regression analysis. Demographic and clinical characteristics of patients were analyzed using an independent 2-sample t -test for interval variables and a χ^2 test for categorical variables. Variables that compared medians were calculated using Mood's median test. Significance was defined as $P < 0.05$. Results are reported as mean (SD), median (interquartile range [IQR]), or number (percentage).

RESULTS

In total, 375 patients were included in the analysis; 282 (75%) were women, and the mean (SD) age was 58 (12) years. Eighty-four patients (22%) had multiple aneurysms treated. The mean (SD) PHASES score for the entire study population was 4.47 (2.67). The mean (SD) length of follow-up was 369 (637) days. The optimal cutoff analysis determined that a PHASES score of ≥ 6 maximized prediction of poor neurologic outcome. Of 375 patients, 116 (31%) had a PHASES score ≥ 6 and 259 (69%) had a PHASES score < 6 .

Patients with PHASES scores ≥ 6 were significantly older than patients with PHASES scores < 6 (63 [11] vs. 56 [11] years, $P < 0.001$) and were significantly more likely to have hypertension (87 [75%] vs. 140 [54%], $P < 0.001$) and a shorter life expectancy ($P < 0.001$; **Table 1**). A greater proportion of patients with PHASES scores ≥ 6 had experienced previous SAH (19 [16%] vs. 20 [7.7%], $P = 0.01$), as well as preoperative anticoagulant and antiplatelet use (44 [38%] vs. 71 [27%], $P = 0.04$) compared with patients with PHASES scores < 6 . mRS score at admission, sex, drug and alcohol use, hypercholesterolemia, hyperlipidemia, and diabetes were not significantly different between the 2 cohorts. Patients with PHASES scores ≥ 6 had significantly higher median (IQR) mRS score at last follow-up (1 [1–2] vs. 1 [1–1]; $P = 0.004$), as well as higher rates of poor neurologic outcome (i.e., mRS score > 2) at discharge (58 [50%] vs. 90 [35%], $P = 0.005$) and follow-up (20 [17%] vs. 18 [6.9%], $P = 0.002$; **Table 2**). After adjusting for optimized cutoffs for age, CCI, nonsaccular aneurysms, and aneurysm size, PHASES score ≥ 6 remained a significant predictor of poor neurologic outcome at follow-up (odds ratio, 2.75; 95% confidence interval, 1.42–5.36, $P = 0.003$; **Table 3**). In fact, PHASES score ≥ 6 was a stronger predictor of the aforementioned confounders. Overall, the PHASES score ≥ 6 threshold alone had a specificity of 71.5%, sensitivity of 52.6%, Youden Index of 0.241, accuracy of 69.6%, and area under the curve of 0.591.

DISCUSSION

With advanced diagnostic imaging has come an increased number of incidentally discovered aneurysms, and additional tools are needed to guide management and predict treatment outcomes for these aneurysms. Here, we provide evidence that the PHASES score could potentially serve as a valuable tool to not only assess the risk of UIA rupture but also predict treatment outcomes. The optimal cutoff analysis determined that a PHASES score of ≥ 6 maximized prediction of neurologic outcome (**Figure 1**). In our study, the percentage of patients with unfavorable neurologic outcomes after a mean follow-up period of 1 year increased from 7% to 17% and thus increased by a factor of approximately 2.5 in patients with PHASES scores ≥ 6 . Moreover, PHASES scores ≥ 6 are associated with an almost three fold increase in the odds of having a poor neurologic outcome after a mean follow-up period of 1 year. Therefore the data in this study showed that a significantly greater proportion of patients with PHASES scores ≥ 6 have a poor neurologic outcome after undergoing surgical treatment for a UIA. Similarly, previous publications have demonstrated that PHASES scores were an independent predictor of poor functional outcomes in patients after aneurysmal SAH.¹² However, because of this study's focus on patients receiving prophylactic care for UIAs,

Table 1. Demographic and Baseline Characteristics in Patients with PHASES (Population, Hypertension, Age, Size, Earlier subarachnoid hemorrhage, Site) Scores <6 and ≥6*

Characteristic	PHASES Score <6, N = 259	PHASES Score ≥6, N = 116	P Value†
Prior SAH	20 (7.7)	19 (16)	0.01
mRS score at admission			0.21
0	73 (28)	24 (21)	
1	146 (56)	68 (59)	
2	40 (15)	24 (21)	
Female sex	196 (76)	86 (74)	0.75
Age, mean (SD), years	56 (11)	63 (11)	<0.001
Family history of SAH or aneurysm	74 (29)	22 (19)	0.049
History of tobacco use	148 (57)	80 (69)	0.03
Current drug abuse	31 (12)	14 (12)	0.98
Current alcohol abuse	4 (1.5)	5 (4.3)	0.14
Life expectancy, based on CCI			<0.001
>10 years	199/254 (78)	62/104 (58)	
5–10 years	55/254 (22)	42/104 (40)	
<5 years	0/254 (0)	2/104 (1.9)	
Diabetes mellitus	36 (14)	22 (19)	0.21
Hypercholesterolemia	74 (29)	38 (33)	0.41
Hyperlipidemia	26 (10)	9 (7.8)	0.48
Anxiety	20 (7.7)	2 (1.7)	0.02
Depression	27 (10)	8 (6.9)	0.28
Other psychiatric conditions	3 (1.2)	4 (3.4)	0.21
Preoperative AC/AP use	71 (27)	44 (38)	0.04
Hypertension	140 (54)	87 (75)	<0.001
Charlson Comorbidity Index			
0	65 (25)	11 (9.5)	
1	53 (20)	19 (16)	
2	74 (29)	26 (22)	
3	43 (17)	29 (25)	
4	18 (6.9)	16 (14)	
5	5 (1.9)	11 (9.5)	
6	1 (0.4)	3 (2.6)	
8	0 (0)	1 (0.9)	

SAH, subarachnoid hemorrhage; mRS, modified Rankin Scale; CCI, Charlson Comorbidity Index; AC, anticoagulant; AP, antiplatelet.

*All values are expressed as no. (%), unless otherwise indicated.

†P values calculated with χ^2 test of independence, Wilcoxon rank-sum test, or Fisher exact test. Boldface type indicates statistical significance.

our findings may help guide clinicians in the decision to treat patients surgically or conservatively after an incidental aneurysm diagnosis.

Other studies evaluating PHASES scores have centered cutoff thresholds around clinical recommendations for treatment and observation of patients with UIAs. In an analysis of 841 patients

with UIAs treated between 2006 and 2014, Bijlenga et al.¹³ attempted to assess the utility of the PHASES score in guiding clinical decision-making and concluded that patients with PHASES score <4 may be considered for observation, whereas a PHASES score of ≥4 indicated a high risk of aneurysm rupture, and intervention was supported for such patients. Juvela¹⁴ found

Table 2. Postoperative Outcomes in Patients with PHASES (Population, Hypertension, Age, Size, Earlier subarachnoid hemorrhage, Site) Scores <6 and ≥6*

Characteristic	PHASES Score <6, N = 259	PHASES Score ≥6, N = 116	P Value†
Residual/recurrent aneurysm	12 (4.6)	9 (7.8)	0.22
Growth/de novo formation on follow-up imaging	37 (14)	21 (18)	0.34
Death	3 (1.2)	3 (2.6)	0.38
mRS score at discharge			<0.001
1	104/258 (40)	25/115 (22)	
2	64/258 (25)	32/115 (28)	
3	86/258 (33)	49/115 (43)	
4	2/258 (0.8)	9/115 (7.8)	
5	2/258 (0.8)	0/115 (0)	
mRS score at discharge >2	90 (35)	58 (50)	0.005
mRS score at final follow-up, median (IQR)	1 (1–1)	1 (1–2)	0.004
mRS score at final follow-up			<0.001
0	46 (18)	25 (22)	
1	155 (60)	48 (41)	
2	40 (15)	23 (20)	
3	13 (5.0)	8 (6.9)	
4	2 (0.8)	8 (6.9)	
5	0 (0)	1 (0.9)	
6	3 (1.2)	3 (2.6)	
mRS score >2 at final follow-up	18 (6.9)	20 (17)	0.002
New neurologic deficit at discharge	70 (27)	45 (39)	0.02
Length of follow-up, mean (SD), days	410 (646)	363 (619)	0.72

mRS, modified Rankin Scale.

*All values are expressed as number (%), unless otherwise indicated.

†Boldface type indicates statistical significance.

Table 3. Multivariate Model Adjustment for PHASES (Population, Hypertension, Age, Size, Earlier subarachnoid hemorrhage, Site) Score ≥6 and Poor Neurologic Outcome at Final Follow-Up

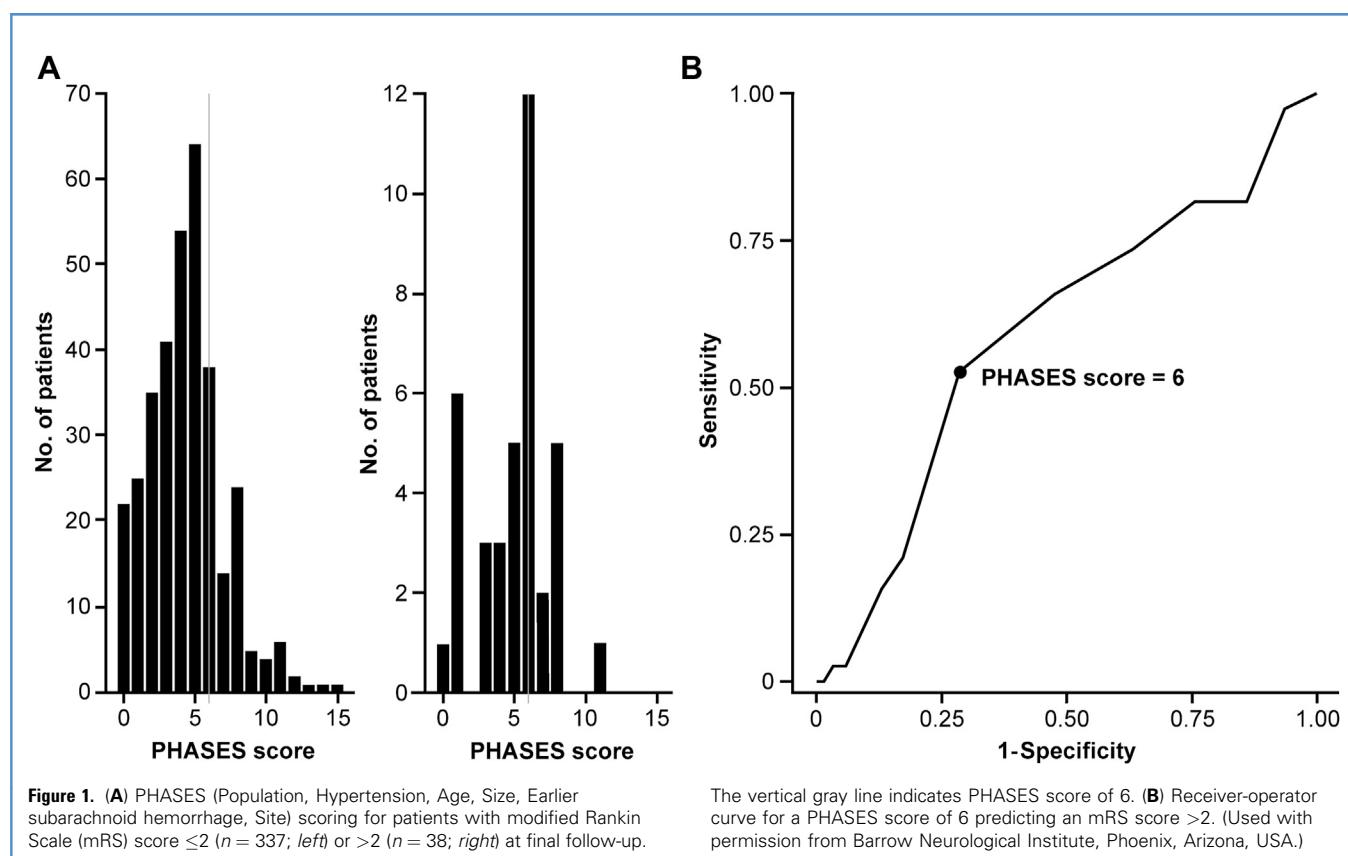
Characteristic	OR	95% CI	P Value*
PHASES score ≥6	2.75	1.42–5.36	0.003
Age ≥61 years	1.25	0.65–2.39	0.50
CCI ≥4	1.08	0.45–2.43	0.86
Nonsaccular aneurysm	1.76	0.55–4.77	0.30
Maximum aneurysm diameter ≥4.96 cm	0.63	0.33–1.22	0.17

OR, odds ratio; CI, confidence interval; CCI, Charlson Comorbidity Index.

*Boldface type indicates statistical significance.

that an optimal cutoff point was ≥6 versus <6 for sensitivity and specificity of predicting aneurysm rupture. However, PHASES score thresholds used in clinical decision-making also appear to differ between treatment centers. Hollands et al.⁹ found that the median PHASES scores ranged between 6 and 8 for patients who underwent treatment at 2 different institutions. Most current PHASES score thresholds solely evaluate the risks associated with patient and aneurysm characteristics; to our knowledge, this study is the first attempt to incorporate treatment outcome in the risk stratification functionality of the PHASES scoring system.

Although the PHASES score was developed to assess aneurysm rupture risk over 5 years, it has been increasingly adopted as a tool to guide management of high-risk unruptured aneurysms in clinical practice.^{8,9} However, use of the PHASES score in clinical decision-making has been controversial because aneurysms that had been deemed low risk on the basis of the PHASES score have ruptured.¹⁵⁻¹⁹ In a 2018 retrospective review of 100 patients



with aneurysmal SAH, Neyzazi et al.¹⁸ found that 70% of aneurysms would have been classified as low risk according to the PHASES score. Similarly, Pagiola et al.¹⁹ noted that more than 78% of their cohort of 155 patients with ruptured aneurysms would not have received PHASES scores that could have accurately predicted their risk of rupture.

Some studies have examined other parameters in an attempt to refine the predicted risk of aneurysm rupture. Patient-related risk factors such as tobacco use,^{20,21} female sex,²² family history of SAH or aneurysm,² collagenopathies,²³ aneurysm shape,²⁴ and changes in aneurysm size²⁵ are not currently accounted for in the PHASES score. With the development of new imaging technologies and analytic techniques, aneurysm-related risk factors for rupture have also been more precisely defined. Liu et al.²⁶ developed a nomogram-based prediction tool that relied on the extraction of quantitative imaging features from radiographic images for aneurysm rupture risk stratification. Their quantitative assessment of aneurysm irregularity—in particular, surface-to-volume ratio and aneurysm flatness—accurately predicted rupture risk and showed high consistency with PHASES scores. Zwarzany et al.²⁷ showed that aneurysm wall enhancement on high-resolution vessel wall magnetic resonance imaging, which is believed to suggest inflammatory changes in the aneurysm wall, correlated with known markers of aneurysm instability. Interestingly, Zwarzany et al.²⁷ also found that aneurysms with a larger enhancement area had significantly higher PHASES scores. Ou

et al.²⁸ used machine learning to develop a model of aneurysm rupture assessment and found that location, size ratio, and triglyceride level were the 3 most important features in predicting rupture. In comparison, Neyazi et al.²⁹ examined 49 morphologic and hemodynamic parameters of aneurysms and found that most of these parameters appeared to have little or no effect on predicting aneurysm rupture in patients with multiple aneurysms. Instead, the authors ultimately used only 2 physiologic parameters, aspect ratio and maximum relative residence time, to generate a model for aneurysm rupture risk. Interestingly, our univariate inclusion model considered many imaging characteristics, such as maximum diameters, perpendicular heights, neck diameters, aspect ratios, calcifications, aneurysm morphology, critically perforating or branching vessels, and intraluminal thrombosis, and none of the characteristics resulted in significance in the univariate or multivariate model before or after thresholding.

Ultimately, other scoring methodologies, including the UCAS (Unruptured Cerebral Aneurysm Study),³⁰ ELAPSS (Earlier subarachnoid hemorrhage, Location of the aneurysm, Age > 60 years, Population, Size of the aneurysm, and Shape of the aneurysm) score,³¹ and UIATS (Unruptured Intracranial Aneurysm Treatment Score),³² have also been developed to predict aneurysm rupture risk and guide clinicians in determining the optimal treatment for patients. The UIATS already considers patient-, aneurysm-, and treatment-related risk

factors and therefore may serve as a superior tool to help determine the appropriate treatment options for patients with UIAs.¹¹ The UIATS is unique in that its components allow for the determination of a score that favors treatment versus observation based on variables that could portend more favorable outcomes given treatment. Although the PHASES score does not allow for this consideration outright, our study allows for a threshold that determines future favorable outcomes given surgery, presenting a potentially more useful prognostication tool. The ELAPSS score attempts to predict aneurysm growth risk, with the overall assumption that a greater growth risk would confer a greater favorability toward treatment. However, the ELAPSS score does not provide insight into favorable outcomes for those receiving surgery, limiting its prognostic value in actual treatment. Given that past research has generated inconclusive results when comparing the predictive capacity of these scores,^{11,16,33,34} the utility of the PHASES scoring system relative to other prognostic tools should be further explored.

Although other arguably superior scoring systems are being increasingly adopted, there is growing evidence questioning the utility of the PHASES score to accurately predict aneurysm rupture risk. With an approximate accuracy of 70% for correctly predicting outcomes, our data support that the PHASES score may, in fact, serve as a valuable tool to predict treatment outcomes, rather than risk of rupture.

Limitations

The main limitation of this study is its retrospective nature and its lack of long-term follow-up data. In general, neither PHASES nor any other scoring system can make completely accurate assumptions and predictions of patient outcomes. Because the patients in our cohort were treated by highly experienced neurosurgeons at a specialized, large-volume quaternary center, the outcome data might not be directly comparable with that for other centers. Only the PHASES scoring system was analyzed in this study, which limits direct comparison with other scoring systems. The PHASES threshold results were significant in the univariate and multivariate models onto patient outcomes; however, evaluation of the score's performance must also consider the strength of predictability, which was limited by a relatively low sensitivity when considered alone.

CONCLUSION

Diagnostic imaging has revealed an increasing number of UIAs, but the decision to treat incidentally discovered aneurysms is challenging. This study demonstrated that PHASES scores ≥ 6 are associated with an approximately threefold increase in the percentage of patients who had a poor neurologic outcome (i.e., mRS score >2) after undergoing surgical treatment for unruptured aneurysms. These findings suggest that PHASES scoring at this threshold may be useful in predicting the treatment outcome for the patient and thus potentially serve as a valuable tool for risk stratification and management for UIAs.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

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